

Laboratory Guide 8: QTris — Quantum Mechanics as a Board Game

A Verified Digital Adaptation of QTris

Course: Quantum Computing Foundation

1 Introduction and Objectives

QTris plays tic-tac-toe on a 3×3 grid whose nine cells are **nine qubits**. Every tile is a quantum state, every move card is a reversible operation, and the endgame is a dice-resolved measurement that collapses the whole board to black and white before a single point can be scored. Its rulebook makes a bold claim: *there is no fundamental difference between QTris and quantum mechanics*. That claim is checkable — and this lab checks it. Every probability in the digital adaptation you will play is computed by an engine whose numbers were verified, state by state and table by table, against the game’s printed measurement tables. Where the print and the physics agree (38 of the 40 printed rows, within a stated rounding of at most half a percentage point), you play the print. Where they provably cannot both be right (two rows — see the erratum section), the physics wins, openly.

You will play first and formalize second. The tally counters, the printed dice tables and an optional exact mode are all part of the instrument: the game is not an analogy for quantum mechanics — it *is* a quantum system you operate by hand, the same physics that runs Stern–Gerlach spin measurements and commercial quantum random-number generators.

Prerequisites:

- *Fundamental Concepts of Quantum Technologies* (lecture series).
- Lab 5: *Quantum Measurements* — the Born rule, measurement Procedures 1 and 2, the states $|0\rangle, |1\rangle, |+\rangle, |-\rangle$ and the four Bell states.
- Lab 6: *Quantum Key Distribution (BB84)* — mutually unbiased bases (this lab’s colour and orientation are exactly that pair).
- Lab 7: *Quantum Optics* — the invariance of $|\Phi^+\rangle$ when both members of a pair are rotated together, which this lab reuses and sharpens.
- Familiarity with Dirac notation: $|\psi\rangle, \langle\phi|, \langle\phi|\psi\rangle$.

Connections to Previous Labs

The Lab 5–7 bridge. The game’s tiles are the Lab 5 states: white and black are the computational basis $|0\rangle, |1\rangle$, the two half-and-half tiles are $|+\rangle, |-\rangle$. Its colour and orientation questions are Lab 6’s mutually unbiased pair, Z versus X . Its entangled pairs are Lab 7’s photon pairs with the analyzers replaced by dice, and the double-pink-square analysis in the erratum section is Lab 7’s “rotate both analyzers together” invariance — now shown to hold for exactly two of the four Bell states.

Mode: Playable Board Game + Qiskit **Credits:** 1 ECTS **Estimated time:** ≈ 8 h

Learning Objectives:

1. **Translate every QTris component into quantum-mechanical language** — tiles as states $|W\rangle = |0\rangle$, $|B\rangle = |1\rangle$, $|R\rangle = |+\rangle$, $|L\rangle = |-\rangle$; cards as unitaries; dice as the Born rule.
2. **Predict measurement statistics** for single tiles and entangled pairs, decorated or not, and confirm them against both the printed d100 tables and the exact probabilities.
3. **Explain why the pink-square gate is a single canonical rotation** $R(-60^\circ)$, and how one rotation produces the game's 25, 75, 93 and 7 — including where the $\sqrt{3}$ comes from.
4. **Distinguish Procedure 1 and Procedure 2** for a rotated-basis measurement: identical statistics, different post-measurement states — on the game's own tiles.
5. **Recognize state-dependent rotational invariance**: why two pink squares cancel exactly on two of the four entangled pair types and provably not on the other two — and how that catches a printed table in an error its own authors missed.
6. **Operate sequential incompatible measurements** (colour vs orientation) and connect them to Stern–Gerlach physics and spin qubits.

About the Original Game

QTris (*QTris e Meccanica Quantistica*) was created by **Alioscia Hamma** (concept and physics) with the creative team **Immacolata De Simone**, **Michela Nazzaro** and **Michele Viscardi** (illustrations by M. Viscardi), produced by AH Quantum Lab and Kommunikart, and published online in 2024 by **NQSTI** — the National Quantum Science and Technology Institute (<https://www.nqsti.it>), Italy's national quantum research partnership. The original rulebook is in Italian and is the authoritative source for every rule and printed table this adaptation implements. This lab is an independent educational adaptation built inside the same national ecosystem, with respect and with every deviation from the printed game declared explicitly (Section 8). The rulebook's own philosophy is worth quoting in spirit: the rules are not a metaphor for quantum mechanics — the rules *are* quantum mechanics. This adaptation takes that philosophy literally enough to verify it. No artwork from the original is reproduced here; only the printed numeric tables are transcribed, for the audit.

The original terms appear throughout the Italian rulebook; this guide uses the English terms below.

Italian (rulebook)	English (this guide)	Meaning
tris	tris (three-in-a-row)	kept as the game’s name element
griglia / casella	grid / cell	the 3×3 board; cells numbered 1–9
tessera	tile	the piece occupying a cell
bianconera	<i>bianconera</i> — the half-and-half tile	the white-and-black tile; kept as a loanword
orientata a destra / sinistra	right- / left-oriented	white half on the right / on the left
carta mossa	move card	a playable operation: I, X, Y, Z, H, U, C_X
tessera quadrata rosa U / rossa H	pink U square / red H square	a marker placed under a tile
decorare	to decorate	placing a square marker under a tile
gattini di Schrödinger	Schrödinger’s kittens	the rulebook’s name for the bianconere
stato entangled	entangled state	one state on two cells (triangle tiles)
fase di preparazione / operazioni / misura	preparation / operations / measurement phase	the three phases of a game
mulligan	mulligan	one optional opening redraw (rulebook’s own loanword)
Espansione	deck preset	Base, $+U$, $+C_X$, $+U+C_X$

Table 1: Translation glossary.

Simulator

The lab’s interactive page ([lab8_qtris.html](#)) hosts the **playable game** built for this lab — hot-seat, two players at one screen — and it is the bench of this guide. Its **Play QTris** tab offers the four official deck presets (**Base**, $+U$, $+C_X$ — the didactic default — and $+U+C_X$), a **dice toggle** between *printed-tables* mode (d100 rolled against the published bands, the consulted row shown on every roll) and *exact* mode (full-precision Born sampling), a seeded random setup or the rulebook’s own example board, an opt-in probability preview, session tallies, and a scoring overlay on the eight lines. Its **Sandbox** tab is a single cell with a free card rack, colour measurement, and orientation measurement by Procedure 1 and by Procedure 2. Every displayed probability flows through the same verified reference engine as the formulas in this guide, and all runs are seeded: the same seed and the same sequence of actions always reproduce the same game.

This guide is written play-first. Each Part opens with a “Play First” box. Do that two-minute run *before* reading the theory: what you see at the board poses exactly the question the section then answers. The companion Qiskit notebook rebuilds the same states as quantum circuits and audits the printed tables in code:

https://colab.research.google.com/github/LMrilo/quantum-labs-2026/blob/main/lab8/lab8_qtris.ipynb

2 Part 1: The Board — Nine Cells, Nine Qubits

Play First

Start a game in the default preset (+ $\mathbf{C}_X$) with the dice toggle on **printed tables**. Play the ten turns any way you like, then run the measurement sweep and watch the dice panel: every roll names the printed row it consults and the band it lands in. Look at the session tally afterwards — the fraction of half-and-half tiles that came up white. Play a few games: it drifts toward **50%**. Keep that number in mind; Part 2 will do something to it that no coin allows.

The Question

Every cell is full from move one, and yet nobody knows who’s winning. What kind of board game can’t tell you its own score until the dice say so?

A game of QTris has three phases. In **preparation**, all nine cells are filled with tiles. In the **operations phase**, the players alternate for ten turns (five each), each turn drawing two cards and playing two — cards act on any cell; there is no cell ownership. In the **measurement phase**, the cells are visited in order $1 \rightarrow 9$ and every unresolved tile is collapsed by a dice roll to plain white or black. Only then is the board scored: one point per three-in-a-row of your colour over the eight lines (three rows, three columns, two diagonals); most points wins.

The tiles are states of a two-level system — a qubit per cell:

$$|W\rangle = |0\rangle, \quad |B\rangle = |1\rangle, \quad |R\rangle = |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |L\rangle = |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \quad (1)$$

White and black are the computational basis. The *bianconera* — half white, half black, with an orientation — is an equal superposition; drawn with the white half on the **right** it is $|+\rangle$ (“right-oriented”), mirrored it is $|-\rangle$. This orientation convention is the one under which every drawing and every passage of the rulebook is self-consistent (one early caption is the sole exception — erratum E3 in Section 7). The colour measurement asks a superposed tile a question it does not have a definite answer to, and the Born rule replies with probabilities:

tile	state	$P(W)$	printed white band (d100)
white	$ 0\rangle$	1	1–100
black	$ 1\rangle$	0	— (no white band)
bianconera, right-oriented	$ +\rangle$	1/2	1–50
bianconera, left-oriented	$ -\rangle$	1/2	1–50

Table 2: The four undecorated tiles and their printed dice rows. A roll of 1–50 on a bianconera is white, 51–100 black.

Formula → Number → Bench

$P(W \mid R) = |\langle 0|+\rangle|^2 = \frac{1}{2}$, i.e. 0.500 — and the printed dice row for the bianconera is exactly 1–50 white, 51–100 black. At the board: measure a few games with bianconere on them and watch the white tally converge to 50%. Then remember that this 50/50 is *not* a coin: Part 2 un-flips it.

Theory Recap

Global phase is discarded — by the game, deliberately. A quantum state and the same state multiplied by a phase factor give identical probabilities for every measurement, so the game identifies them. Two consequences you can check on the move cards: the card Z does nothing visible to a monochrome tile ($Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle$) and the card X does nothing visible to a bianconera ($X|\pm\rangle = \pm|\pm\rangle$), so both are *legal but invisible* plays — the digital game lets you make them and tells you so. The card Y acts as “ X and Z at once”: $Y = iXZ$, and the i is exactly what the game discards. Even the card I — do nothing — is a legal tempo move. A move that provably does nothing is itself teachable: it is what *global phase* means.

Real-World Anchor

A research institute’s outreach instrument. QTris is not a classroom toy invented for this course. It is an instrument of Italy’s national quantum initiative: conceived by a physicist, produced by a design team, playtested by university researchers and published by NQSTI — the same ecosystem this course belongs to. Game-based learning is an active research line in quantum education (see the survey by Seskir et al. in the Resources), and QTris sits at its most literal end: the rules are not a metaphor for the theory, they *are* the theory, which is why an engine can audit them.

3 Part 2: The Bianconera — Why Orientation Matters

Play First

In any preset, find a right-oriented bianconera on the board (or make one: play H on a white tile). Now play H on it and run the sweep. The tile needs no dice at all — it is white, with certainty. Do it with a left-oriented one: black, with certainty. The rulebook itself recommends this tactic.

The Question

A half-white-half-black tile turns up white exactly half the time. So does a coin. Why does the game bother printing an orientation on the tile — what could the direction of a 50/50 tile possibly change?

The move cards answer it. The card H connects the two families: it turns white into a right-oriented bianconera and back, black into a left-oriented one and back. So play H on a bianconera and the superposition *closes*:

$$H|+\rangle = |0\rangle, \quad H|-\rangle = |1\rangle. \quad (2)$$

Counterintuitive — and Real

One card un-flips the coin. No operation on a fair coin can make its next toss certain — a probability distribution has no handle to grab. But H on a right-oriented bianconera yields a *certainly white* tile: 100%, not 50. The bianconera was never a distribution; it is a state with a definite *orientation*, and the orientation is the handle. The rulebook calls these tiles Schrödinger’s kittens for exactly this reason: the superposition is a real, manipulable thing.

The same logic exposes the phase. The card Z flips orientation, $Z|\pm\rangle = |\mp\rangle$, yet changes *nothing* a colour measurement can see — both orientations measure 50/50. Invisible? Only until an H converts the hidden difference into opposite certainties: $H|+\rangle = |0\rangle$ (white for sure) while $H|-\rangle = |1\rangle$ (black for sure).

Counterintuitive — and Real

The Z card changes nothing you can see — until it changes everything. Phase is not bookkeeping. The game tracks it with tile orientation, and one H card cashes it out as opposite certain colours. This is the same relative phase that Lab 7's interferometer converted into which-detector certainty.

Formula $\rightarrow$ Number $\rightarrow$ Bench

$H|+\rangle = |0\rangle$, so $P(W \mid H \text{ on } R) = 1$: the number is 1.000 — not 0.500. At the board: play H on a bianconera, then measure. This is how you build a guaranteed tris piece out of chance.

Deep Dive

The single-tile card algebra. The rulebook's operation map for single tiles is the Pauli/Hadamard action on the dictionary of equation (1), taken modulo global phase: X is the colour flip ($X|0\rangle = |1\rangle$; invisible on bianconere), Z is the orientation flip ($Z|\pm\rangle = |\mp\rangle$; invisible on monochrome tiles), Y does both at once ($Y = iXZ$, phase discarded), and H is the two bridges $W \leftrightarrow R$, $B \leftrightarrow L$. Every one of these cards is its own inverse: $X^2 = Y^2 = Z^2 = H^2 = I$, so any card played twice on the same tile is a round trip — the rulebook's own unitarity note, made tactile. Playing H twice on a bianconera therefore returns the bianconera; playing Z between the two H 's returns the *opposite* colour — phase, converted twice.

Connections to Previous Labs

Lab 7 converted a relative phase between two paths into a certain detector click at the second beam splitter. Equation (2) is the same conversion with the beam splitter replaced by an H card and the two paths by the two colours: the orientation of a bianconera is a phase, and H is the device that reads it.

4 Part 3: Entanglement — One State on Two Cells

Play First

In preset $+C_X$, play the C_X card with a bianconera as control and a monochrome tile as target (the game asks you to declare which is which). The two cells become a pair of triangles — one state on two cells. Run the sweep, and watch the pair counter in the session tally over several games: **100% agreement**, while each member's own colour tally sits at 50/50. Entangle cells 1 and 9 — opposite corners — and check that distance changes nothing.

The Question

Two triangles on opposite corners of the board always come up the same colour, every game, though each alone is a perfect coin toss. Where does the agreement live?

The two-cell card C_X (a controlled-NOT: the player declares which cell is control and which is target) is the game's only source of correlation. Played with a bianconera as control and a monochrome tile as target, it *entangles* them into one of the four Bell states, shown on the board as a pair of triangles:

$$(R, W) \rightarrow \Phi^+, \quad (R, B) \rightarrow \Psi^+, \quad (L, W) \rightarrow \Phi^-, \quad (L, B) \rightarrow \Psi^- \quad (3)$$

— control orientation $\mapsto$ sign of the pair, target colour $\mapsto$ same-colour (Φ) or opposite-colour (Ψ) type. The recipe is one line of algebra: with the control written first,

$$|+\rangle|0\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{C_X} \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |\Phi^+\rangle, \quad (4)$$

and likewise $|+\rangle|1\rangle \rightarrow (|01\rangle + |10\rangle)/\sqrt{2} = |\Psi^+\rangle$, $|-\rangle|0\rangle \rightarrow (|00\rangle - |11\rangle)/\sqrt{2} = |\Phi^-\rangle$, $|-\rangle|1\rangle \rightarrow (|01\rangle - |10\rangle)/\sqrt{2} = |\Psi^-\rangle$.

Reading order (used everywhere in this guide, on the interactive page and in the notebook): pair outcomes like WB are written **big-endian** — the lower-numbered cell is the **leftmost** letter, exactly as $|q_0q_1\rangle$ puts qubit 0 leftmost. Amplitude vectors are listed in the order $(WW, WB, BW, BB) = (|00\rangle, |01\rangle, |10\rangle, |11\rangle)$. The companion notebook shows how to convert the opposite ordering used by common circuit software.

A Φ^+ pair measures WW or BB — never mixed — each with probability $\frac{1}{2}$: each member alone is perfectly random, the two together perfectly agreed. The agreement does not live in either cell; it lives in the pair. Adjacency is irrelevant — cells 1 and 9 entangle as happily as neighbours. And the same card played again on the pair *disentangles* it back into the tiles it came from (the player chooses which output tile lands on which cell): every operation in the game is reversible; only the dice are not.

Counterintuitive — and Real

Two cells that never touched agree forever. One C_X card creates perfect correlation with individually perfect randomness, at any board distance. This is the same resource measured in Lab 7's entangled-photon correlations, and the resource behind quantum key distribution (Lab 6): the correlation is a property of the joint state, stored in neither place.

Formula $\rightarrow$ Number $\rightarrow$ Bench

$P(\text{same colour} \mid \Phi^+) = |\langle 00 | \Phi^+ \rangle|^2 + |\langle 11 | \Phi^+ \rangle|^2 = \frac{1}{2} + \frac{1}{2} = 1$: agreement 1.000, and the mixed outcomes have probability exactly 0 — bands that simply do not exist on the printed row (1–50 $\rightarrow WW$, 51–100 $\rightarrow BB$, nothing else). At the board (preset $+C_X$): entangle two cells, measure many games, and watch the pair counter pin agreement at 100% while each member's own tally sits at 50/50.

Deep Dive

The marker calculus — how decorations compose. A pair member may carry markers under its triangle: a red H square (the H card played on that member) and a pink U square (Part 4). Each member's stack is one of $\{I, H, U, UH\}$, read from the tile outward, with U always outermost: the H card toggles the layer next to the tile, the U card toggles the top layer. Three identities make the printed operation tables come out exactly:

- **Both members marked H normalize away.** $(H \otimes H)$ leaves $|\Phi^+\rangle$ and $|\Psi^-\rangle$ unchanged and *swaps* the other two: $(H \otimes H)|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|++\rangle - |--\rangle) = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) = |\Psi^+\rangle$. So when the second member receives an H marker too, both markers come off and the pair is relabelled $\Phi^- \leftrightarrow \Psi^+$ — the rulebook prints exactly this column.
- **Paulis commute through H with $X \leftrightarrow Z$.** $HX = ZH$ and $HZ = XH$ (and Y passes through up to phase). A Z card on an H -marked member therefore acts as X on the core — which is why the rulebook needs *two* X columns and *two* Z columns for marked pairs (played on the marked vs the unmarked member) but only one Y column.
- **Phase kickback.** C_X with a bianconera control *and* a bianconera target flips the *control's* orientation exactly when the target is left-oriented: $C_X|+\rangle|-\rangle = |-\rangle|-\rangle$, while a right-oriented target is an eigenstate and the play is invisible. The X -basis CNOT runs backwards — the case the printed tables do not cover, fixed by the physics.

Counting the state space. 8 single-cell states (4 tiles $\times$ with/without pink square) and $4 \times 12 = 48$ entangled states (4 Bell cores $\times$ the $4 \times 4 = 16$ stack pairs minus the 4 in which both members carry H , which normalize away): 56 states in all, closed under every card. Small enough that every claim in this guide was checked exhaustively, not sampled.

Theory Recap

No signalling, printed. Whatever markers sit under *one* member of a pair, the *other* member alone is a perfect coin: both single-cell marginals of every pair row are exactly $\frac{1}{2}$, because local operations cannot change the reduced state $I/2$ of a Bell pair (Lab 7's "each photon alone is a perfect coin"). The printed tables respect this *even after rounding*: where the exact eighths 12.5/37.5 had to become integers, the authors rounded them in opposite directions (12 and 38) so that every row still tiles 1–100 and both marginals still sum to exactly 50. Table 5 shows the result.

Connections to Previous Labs

Lab 7's Part D showed photon pairs agreeing at matched analyzer angles while each side alone stayed 50/50; Lab 6 used the same correlations as the raw material of a key. The C_X card is the pair source, the sweep is the pair of analyzers, and the printed Φ^+ row — WW or BB , nothing else — is the matched-angle perfect correlation.

5 Part 4: The Pink Square — One Hidden Rotation (advanced)

Play First

Switch to preset **+U**. Play the U card on a right-oriented bianconera — a pink square appears under it — and run the sweep in printed-tables mode: the dice panel consults a **1–93** row. Flip the toggle to **exact** mode and measure again: the panel now serves 0.933013 with all its digits. Decorate a white tile instead: **1–25**.

The Question

One pink square turns your certain white tile into a 25% underdog. What single operation produces 25, 75, 93 *and* 7 — and why does the 93 come with a $\sqrt{3}$ in it?

The rulebook never writes the pink square’s matrix — it only prints its measurement table: a decorated white tile measures white 25%, decorated black 75%, decorated bianconere 93% and 7%. Those four rows pin the gate completely. Take the square to be a canonical (real) rotation by an angle θ ,

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad R(\theta)|0\rangle = \cos \theta |0\rangle + \sin \theta |1\rangle. \quad (5)$$

The decorated-white row demands $\cos^2 \theta = \frac{1}{4}$, hence $\theta = \pm 60^\circ$; matching the 93 to the *right*-oriented bianconera (and the 7 to the left-oriented one) fixes the sense:

$$U = R(-60^\circ) = \begin{pmatrix} 1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{pmatrix}. \quad (6)$$

Now $|+\rangle = R(45^\circ)|0\rangle$, and rotations compose by adding angles, so

$$U|+\rangle = R(-60^\circ)R(45^\circ)|0\rangle = R(-15^\circ)|0\rangle, \quad (7)$$

$$P(W \mid U \text{ on } R) = \cos^2 15^\circ = \frac{1 + \cos 30^\circ}{2} = \frac{2 + \sqrt{3}}{4} \approx 0.9330. \quad (8)$$

The game’s mysterious 93 is $\cos^2$ of fifteen degrees — an irrational probability, printed to two digits. Table 3 lists all eight single-cell rows against the print.

tile	state	exact $P(W)$	printed white band	gap
white	$ 0\rangle$	1	1–100	0
black	$ 1\rangle$	0	—	0
bianconera right	$ +\rangle$	$1/2$	1–50	0
bianconera left	$ -\rangle$	$1/2$	1–50	0
white + U	$U 0\rangle$	$1/4$	1–25	0
black + U	$U 1\rangle$	$3/4$	1–75	0
bianconera right + U	$U +\rangle = R(-15^\circ) 0\rangle$	$(2 + \sqrt{3})/4 \approx 0.9330$	1–93	0.0030
bianconera left + U	$U -\rangle$	$(2 - \sqrt{3})/4 \approx 0.0670$	1–7	0.0030

Table 3: The eight single-cell states and their printed dice rows. The two irrational rows are rounded by the print to two digits; the gap is 0.0030.

Unlike X , Y , Z and H , this rotation does not map the simple tile family onto itself ($U|0\rangle$ is none of the four tiles) — which is exactly why the pink square must *stay on the board* as a marker instead of being absorbed into a relabelled tile, and why the + U presets are the advanced tier: one little square carries the game beyond the flip-and-mix world into genuinely new probabilities.

In the language of Lab 5, U is not a Clifford operation: it does not send Pauli operators to Pauli operators.

Formula → Number → Bench

$P(W \mid U \text{ on } R) = \cos^2 15^\circ = \frac{2+\sqrt{3}}{4}$. As a float64 number this is 0.9330127018922193 (exact mode displays 0.933013); the print's 93 rounds the physics by 0.0030, within its stated half-point tolerance. At the board (preset +U): decorate a bianconera, roll against the printed 93/7 row, then flip to exact mode and see the value served with all its digits.

Deep Dive

Where the stack order is pinned, and where the eighths come from. Two facts about decorated *pairs* follow from equation (6).

- **U outermost, not innermost.** A member carrying both a red and a pink square could mean UH (apply H , then U) or HU . The two differ: $|\langle 0|UH|0\rangle|^2 = |\langle 0|U|+\rangle|^2 = \frac{2+\sqrt{3}}{4}$ while $|\langle 0|HU|0\rangle|^2 = \frac{2-\sqrt{3}}{4}$. On a Φ^+ pair the first gives the outcome probabilities (q_+, q_-, q_-, q_+) with $q_\pm = \frac{2\pm\sqrt{3}}{8} \approx 0.4665, 0.0335$ — the printed **47/3/3/47** row — and the second gives (q_-, q_+, q_+, q_-) , i.e. 3/47/47/3. The print decides: the pink square sits outermost.
- **The eighth family.** With a pink square on member 1 only, $(U \otimes I)|\Phi^+\rangle = \frac{1}{\sqrt{2}}(U|0\rangle|0\rangle + U|1\rangle|1\rangle)$, and $U|0\rangle = (\frac{1}{2}, -\frac{\sqrt{3}}{2})$, $U|1\rangle = (\frac{\sqrt{3}}{2}, \frac{1}{2})$ give the amplitude vector $\frac{1}{2\sqrt{2}}(1, \sqrt{3}, -\sqrt{3}, 1)$ over (WW, WB, BW, BB) , hence the probabilities $(\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8})$ — the rows the rulebook resolves exactly with an eight-sided die.
- **Mirror symmetry.** Swapping the two members' stacks swaps the WB and BW columns and nothing else, which is why the rulebook may tabulate marker-on-member-1 configurations without loss.

pattern	exact (WW, WB, BW, BB)	printed d100 bands	exact die
corr	$(\frac{1}{2}, 0, 0, \frac{1}{2})$	1-50 / — / — / 51-100	
anti	$(0, \frac{1}{2}, \frac{1}{2}, 0)$	— / 1-50 / 51-100 / —	
unif	$(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$	1-25 / 26-50 / 51-75 / 76-100	
e^8	$(\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8})$	1-12 / 13-50 / 51-88 / 89-100	d8: 1 / 2-4 / 5-7 / 8
$\bar{e}^8$	$(\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{3}{8})$	1-38 / 39-50 / 51-62 / 63-100	d8: 1-3 / 4 / 5 / 6-8
g	(q_+, q_-, q_-, q_+)	1-47 / 48-50 / 51-53 / 54-100	
$\bar{g}$	(q_-, q_+, q_+, q_-)	1-3 / 4-50 / 51-97 / 98-100	

Table 4: The seven outcome patterns of the printed pair tables, with $q_\pm = (2 \pm \sqrt{3})/8$. The rulebook's footnotes give the eight-sided-die procedure for the eighth-family rows and call their d100 bands an approximation.

member 1	member 2	Φ^+	Ψ^+	Φ^-	Ψ^-
I	I	corr	anti	corr	anti
H	I	unif	unif	unif	unif
U	I	e^8	$\bar{e}^8$	e^8	$\bar{e}^8$
UH	I	g	$\bar{g}$	g	$\bar{g}$
I	U	e^8	$\bar{e}^8$	e^8	$\bar{e}^8$
H	U	g	$\bar{g}$	$\bar{g}$	$\bar{g}$
U	U	corr	$\bar{e}^{8\dagger}$	$e^{8\dagger}$	anti
UH	U	$\bar{g}$	unif	unif	g

Table 5: The 32 printed pair rows (8 marker configurations $\times$ 4 Bell cores), every entry the exact Born distribution of the decorated state. All 32 printed rows match these patterns within the rounding of Table 4 — except the two marked † , which the rulebook prints as anti and corr respectively: erratum E2, Section 7. Note the (H, U) row grouping by *sign* rather than by colour type — non-obvious, and exactly right.

Important

The rounding bound, audited. Every printed d100 band is an integer, so the exact values $\{0, 1, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \frac{3}{8}, p_\pm, q_\pm\}$ appear as $\{0, 100, 50, 25, 75, 12, 38, 93, 7, 47, 3\}$ out of 100. Over all 38 conforming rows (8 single-cell + 30 pair rows) the largest gap between print and physics is **exactly** 0.005 — half a percentage point, attained by the $12.5 \rightarrow 12$ and $37.5 \rightarrow 38$ roundings; the irrational rows err by 0.0030 (93/7) and 0.0035 (47/3). For the eighth-family rows the rulebook itself supplies an exact eight-sided-die procedure and calls the d100 version an approximation — the source documents its own rounding layer. The digital game’s exact mode extends that exactness to every row, including the irrational ones no die can realize.

Real-World Anchor

Randomness as a resource. The measurement phase is a hand-executed Born sampler — the same physics that commercial quantum random-number generators package in a chip: a superposed state, a projective measurement, an outcome no algorithm predicts. The dice toggle is that industry’s central distinction made playable: printed-tables mode is a fixed integer approximation of the distribution (a pseudo-randomness of the values), exact mode samples the true Born probabilities. On 38 of 40 rows you cannot tell them apart to better than half a percentage point; on the two erratum rows only the physics is served.

6 Part 5: Measurement — The Only Move You Can’t Take Back

Play First

Open the **Sandbox** tab. Reset the cell to W and press the guided chain: *orientation* $\rightarrow$ *colour* $\rightarrow$ *orientation*. Run it many times and read the counters: **50/50 at every step**. The white tile you started from is gone after the first orientation measurement — and measuring the colour again does not bring it back. Then run both orientation procedures on the same decorated tile and compare what is left on the cell afterwards.

The Question

You just measured the tile white. Measure its orientation, then look at the colour again: half your whites turned black. Who repainted the tile?

The endgame sweep visits cells $1 \rightarrow 9$. Monochrome tiles are skipped; every superposed or decorated tile resolves by one roll against its printed row; a triangle pair resolves *jointly* — one roll for both cells — when the sweep first reaches it (at its lower-numbered cell; the partner is monochrome by the time the sweep arrives there). Every resolved state is replaced by a plain, undecorated outcome tile. Sweep again and nothing changes and no dice are consumed: that is collapse, operationally — outcomes, once real, are stable.

Counterintuitive — and Real

Measurement is the game’s only irreversible move. Every card can be undone — X , Y , Z , H and the pink square are their own inverses, C_X un-entangles what it entangled. The dice cannot be un-rolled, and a white tile carries no memory of the bianconera it came from. Reversible dynamics, irreversible measurement: the deepest asymmetry in quantum mechanics, playable with cardboard.

The **sandbox** lifts the rulebook’s single-cell solitaire into a lab instrument. A colour measurement asks the $|0\rangle/|1\rangle$ question. An *orientation* measurement asks the $|+\rangle/|-\rangle$ question, and the game offers it in two equivalent-but-different ways, named as in Lab 5.

Procedure 1 vs Procedure 2

Two ways to ask the orientation question.

- **Procedure 1** projects directly onto the orientation states $\{|+\rangle, |-\rangle\}$: probabilities $|\langle\pm|\psi\rangle|^2$, and the post-measurement state is an oriented bianconera tile.
- **Procedure 2** is the rulebook’s own construction: apply H — the canonical rotation for this basis — measure *colour*, and relabel the outcomes (“swap the labels on the apparatus”). The post-measurement state is a monochrome tile.

Identical statistics, different post-states. The statistics agree because $|\langle 0|H|\psi\rangle|^2 = |\langle +|\psi\rangle|^2$ for every ψ — checked in the sandbox on a state where the answer is *not* 50/50: for the decorated tile $U|+\rangle = R(-15^\circ)|0\rangle$ both procedures give $P(+)=\frac{1}{4}$, $P(-)=\frac{3}{4}$ (since $\cos 15^\circ - \sin 15^\circ = \sqrt{1 - \sin 30^\circ}$). The post-states differ: the overlap between Procedure 1’s $|+\rangle$ tile and Procedure 2’s $|0\rangle$ tile is $|\langle +|0\rangle|^2 = \frac{1}{2}$, not 1. Downstream cards see different tiles — the Lab 5 and Lab 7 distinction, now on cardboard.

Colour and orientation are *incompatible*: measuring one completely randomizes the other. Starting from a white tile, the chain orientation $\rightarrow$ colour $\rightarrow$ orientation gives 50/50 at every step. Nobody repainted the tile — the colour question simply has no answer while the orientation is definite, and vice versa.

Formula → Number → Bench

$|\langle 0|+\rangle|^2 = |\langle +|0\rangle|^2 = \frac{1}{2}$ — the two bases are mutually unbiased (Lab 6’s key-distribution pair, in tile form), so the chain reads 0.500 at every step, forever. In the sandbox: run the guided chain from W ; then run both procedures on the same decorated tile and compare post-states. And for the scoring rule, three boards you can set up and verify: an all-white board scores 8–0 (every line); white on the odd cells 1, 3, 5, 7, 9 and black elsewhere scores 2–0 (both diagonals); a white top row over a black remainder scores 1–2.

Real-World Anchor

Stern–Gerlach as a solitaire. Replace colour/orientation by spin- z /spin- x and the sandbox chain is the sequential Stern–Gerlach experiment: a spin-up beam, measured along x , then along z again, is 50/50 at every stage. That is exactly the readout principle of today’s spin-qubit hardware, which reads one electron’s spin at a time and pays the same price — a measurement along one axis erases the other. The game’s single cell is the same physics with dice.

Connections to Previous Labs

Lab 6 needed two bases that reveal nothing about each other — Z and X — so that an eavesdropper measuring in the wrong one both learns nothing and leaves a trace. The sandbox chain is that property on one tile: colour and orientation are exactly those two bases, and “half your whites turned black” is the trace.

7 The Erratum Section: When Formalism Referees the Referee

This adaptation verified every printed measurement row against the underlying quantum mechanics. The result honours the game: **38 of the 40 printed rows are exact within their own stated rounding** — the largest gap is half a percentage point, and the authors even rounded 12.5/37.5 in opposite directions (12 and 38) so the row totals and marginals stay perfect. Three small slips remain, declared here openly, because the game’s own philosophy — the rules *are* the physics — deserves to be taken seriously.

Counterintuitive — and Real

The printed game itself slipped — and the formalism catches it. The double-pink-square table claims two pink squares never matter on an entangled pair. That is exactly true for two of the four pair types — and provably impossible for the other two. Rotational invariance is a property of *specific states*, not of entanglement in general. An engine that checks every number caught a table that playtesting could not.

Erratum

E2 — the double- U rows. The printed table for pairs with a pink square on *each* member (the rulebook’s Fig. 2.16, top) shows all four pair types unchanged — perfect correlation or anticorrelation, as if the squares cancelled. Here is the derivation, step by step.

Step 1 — the matrix picture. Write a two-cell amplitude vector as a 2×2 matrix M whose entry M_{jk} is the amplitude of $|jk\rangle$. Then $|\Phi^+\rangle \leftrightarrow I/\sqrt{2}$, $|\Phi^-\rangle \leftrightarrow Z/\sqrt{2}$, $|\Psi^+\rangle \leftrightarrow X/\sqrt{2}$, $|\Psi^-\rangle \leftrightarrow ZX/\sqrt{2}$, and a product operation acts as

$$(A \otimes B)|\Phi^+\rangle \leftrightarrow AB^T/\sqrt{2}, \quad \text{so} \quad (U \otimes U)|\Phi^-\rangle \leftrightarrow UZU^T/\sqrt{2}. \quad (9)$$

Step 2 — the two rows that are right. $UU^T = I$ for any rotation, so $(U \otimes U)|\Phi^+\rangle = |\Phi^+\rangle$ exactly (Lab 7’s “rotate both together” invariance). And $U(ZX)U^T = ZX$ for every rotation, because ZX is itself the rotation by 90° and rotations commute — so the singlet $|\Psi^-\rangle$ is invariant too. Rows 1 and 4 of the printed table are exactly right.

Step 3 — the two rows that cannot be. For the printed Φ^- row (perfect correlation) the matrix UZU^T would have to be *diagonal*. But for $U = R(\theta)$,

$$R(\theta)Z R(\theta)^T = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}, \quad \theta = -60^\circ: \quad UZU^T = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}, \quad (10)$$

so the off-diagonal amplitude is $-\frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}}$ and

$$|(UZU^T)_{01}|^2 = \sin^2 120^\circ = \frac{3}{4} \neq 0. \quad (11)$$

The exact distribution of the double- U Φ^- pair is therefore $(\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8})$ over (WW, WB, BW, BB) ; the same computation with X in place of Z gives $(\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{3}{8})$ for the double- U Ψ^+ pair. Against the printed $(\frac{1}{2}, 0, 0, \frac{1}{2})$ and $(0, \frac{1}{2}, \frac{1}{2}, 0)$ that is a deviation of exactly $\frac{3}{8}$ in every outcome — not a rounding of anything.

Step 4 — no other gate would help. Suppose *some* unitary U made both the printed Φ^+ row and the printed Φ^- row true. Then UU^T and UZU^T would both be diagonal, hence so would their average $U \frac{I+Z}{2} U^T = U|0\rangle\langle 0|U^T = uu^T$ with $u = U|0\rangle$. Its off-diagonal entry is u_0u_1 , so $u_0u_1 = 0$ and $|\langle 0|U|0\rangle|^2 \in \{0, 1\}$ — flatly contradicting the printed 25/75 row for a decorated white tile. The same argument rules out any *pair* of gates $U \otimes V$. No assignment of any rotation, indeed any unitary, to the pink square can make the printed table true.

Resolution in this adaptation. The two rows resolve by the exact eighth-partitions on an eight-sided die — Ψ^+ : 1–3 / 4 / 5 / 6–8, Φ^- : 1 / 2–4 / 5–7 / 8 — in *both* dice modes (the same exact-die procedure the rulebook itself provides for its other eighth-valued rows); the printed bands for these two rows are shown only in the interactive page’s erratum panel, as documentation. Framing matters: this is a transcription slip in an otherwise remarkably accurate physical game — two rows out of forty, in the hardest table of the book. The design is sound; the formalism simply referees everyone, authors included. That is the point of formalism.

Erratum

E1 — one number in a margin note. A collapse-stability note late in the rulebook (p. 37) quotes probability $\frac{1}{4}$ for a decorated right-oriented bianconera turning up white. The book’s own printed table (the 1–93 row) and the physics give $\frac{2+\sqrt{3}}{4} \approx 0.933$; the quoted $\frac{1}{4}$ belongs to the decorated *white* tile two rows up. An authorial slip with no effect on any rule — the note’s actual argument (collapsed outcomes are stable) is untouched. Recorded so nobody re-quotes the number.

Erratum

E3 — two swapped words in one caption. One early figure caption (the rulebook’s Fig. 2.3) names its right-oriented cell “left-oriented” and vice versa — the words are swapped relative to the drawing. Every other orientation-bearing figure and passage in the book (five drawings, four independent prose passages) agrees with the drawing, so the drawing is authoritative and this adaptation’s tiles follow it: right-oriented means *white half on the right*. Worth recording because readers comparing this guide against that one caption would otherwise think the adaptation had it backwards. No probability is affected.

8 Declared Deviations from the Printed Game

Everything this adaptation does differently from the physical edition, in one list:

1. **Ambiguity resolutions.** Where the printed rules are silent — the exact mulligan procedure, tie handling, who is control and who is target when C_X is played, how stacked markers compose, how the d100 is read, the three-pair limit — this adaptation fixes each gap physics-first, guided by the rulebook’s own examples, and enforces the result consistently. Notably: the player *declares* control and target on every C_X ; marker stacks compose with the pink square outermost (the unique order matching the printed 47/3 rows); ties re-roll; equal scores are a draw.
2. **Exact mode.** The physical game rolls d100 against printed integer bands. This adaptation adds an optional exact mode that samples the true probabilities in full precision — and documents that on 38 of 40 rows the printed bands sit within 0.005 of the truth. The printed-tables mode remains the default and reproduces the published bands verbatim.
3. **English translation.** The original is Italian; this guide and the interactive page are English, with the glossary of Table 1. Rule semantics are unchanged in translation.
4. **A seam for a computer opponent.** The engine is a pure state machine with no assumptions about who chooses moves, so a vs-computer mode can attach later. It is deliberately *not* implemented: hot-seat play between two humans is the mode the original game is designed around.
5. **The two erratum rows are not playable as printed.** The double- U rows for Ψ^+ and Φ^- resolve by the exact eighth-partitions in *both* dice modes; the impossible printed bands appear only in the erratum panel. Serving the physics — openly — was judged more faithful to the game’s philosophy than serving the typo.

For completeness, the game’s own numbers as implemented: the four deck presets are Base (I 6, X 12, Y 6, Z 14, H 18), + U (5, 10, 5, 11, 11, U 10), + C_X (5, 10, 5, 11, 11, C_X 10) and + $U+C_X$ (4, 8, 4, 9, 9, U 9, C_X 9); six cards are dealt to each player, at most one mulligan

each, then ten turns of draw two / play two, so the deck is never exhausted ($12 + 20 + 12 = 44$ cards at most, counting a worst-case mulligan, against the smallest preset); turn order and the colour White go to the higher of two eight-sided dice; at most three entangled pairs coexist on the board.

9 Part 6: Guided Activities

Simulator

All activities below run on the game and sandbox of the lab's interactive page ([lab8_qtris.html](#)); the preset to use is named in each activity. All runs are seeded — the same seed always reproduces the same game. Activity 6 uses the companion Qiskit notebook:

https://colab.research.google.com/github/LMrilo/quantum-labs-2026/blob/main/lab8/lab8_qtris.ipynb

9.1 Activity 1: The Bianconera Tally and the H Rescue (Preset $+C_X$)

Student Activity

1. Play several games in printed-tables mode and read the session tally: what fraction of half-and-half tiles came up white? Explain the number with equation (1) and the Born rule.
2. In the operations phase, play H on a right-oriented bianconera and run the sweep. Why does this tile consume no dice roll? Which state was it in before, and which after? (Equation (2).)
3. Play Z on a bianconera, then H , then measure. Compare with H alone. Where was the difference stored while the tile still measured 50/50?

9.2 Activity 2: The Pair Counter (Preset $+C_X$)

Student Activity

1. Entangle two far-apart cells with C_X (bianconera control, monochrome target). Over several games, verify that the pair counter reads 100% agreement while each member's own tally stays near 50/50.
2. Using the recipe of equation (3), prepare each of the four Bell pairs from tiles and check, with the probability preview, which two give WW/BB and which two give WB/BW . Why is the sign of the pair invisible to the colour measurement?
3. Play C_X again on the pair you made. What comes back, and who chooses where? Starting from a white and a black tile, prepare the Ψ^- pair using only cards — one solution uses two cards.

9.3 Activity 3: The 93 and Exact Mode (Preset +U)

Student Activity

1. Decorate a right-oriented bianconera with the pink square and measure it in printed-tables mode: name the row consulted. Switch to exact mode and measure again: read the probability the panel serves and compare with equation (8).
2. Decorate a white tile, then a black tile. Verify 25% and 75% against Table 3, and explain both from the matrix (6).
3. Why can no card stack ever make a pink-decorated tile *certain*? (Its four possible values of $P(W)$ are $\frac{1}{4}$, $\frac{3}{4}$ and $(2 \pm \sqrt{3})/4$.) What single card removes the problem?

9.4 Activity 4: Making the Erratum Visible (Preset +U+C_X)

Student Activity

1. Build a Φ^- pair (left-oriented control on a white target), decorate *both* members with pink squares, and measure many times in either dice mode. Compare your tallies with the exact eighths $(\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8})$ and with the printed row shown in the erratum panel.
2. Do the same with a Φ^+ pair. Why do two pink squares cancel here and not on Φ^- ? Point to the line of Section 7 that decides it.
3. State in one sentence why *no* choice of gate for the pink square could make the printed double- U table right (Step 4 of erratum E2).

9.5 Activity 5: The Sandbox Chain and the Two Procedures

Student Activity

1. Run the guided chain $W \rightarrow \text{orientation} \rightarrow \text{colour} \rightarrow \text{orientation}$ many times. Record the counters at each step and explain the three 50/50 results with mutual unbiasedness.
2. On the same decorated tile $U|+\rangle$, run Procedure 1 and Procedure 2 many times each. Confirm the statistics agree (and are not 50/50). Then compare the tiles left on the cell afterwards.
3. Try to find a state for which Procedures 1 and 2 give different *statistics*. (You can't — that is the theorem. Then say precisely what *does* differ.)

9.6 Activity 6: The Game in Qiskit

The companion notebook rebuilds this guide's states as quantum circuits: the eight single-cell tiles (the pink square as an explicit $R(-60^\circ)$, also written as a standard R_y gate), the four Bell pairs prepared exactly the way the C_X card prepares them, the full printed measurement table reproduced from state-vector probabilities — including a computed, not asserted, verdict on the two erratum rows — seeded sampling runs beside the exact numbers, and the two orientation procedures side by side. Its tile-preparation circuit is a few lines long:

```
1 def single_cell_circuit(name: str) -> QuantumCircuit:
```

```

2  """Preparation circuit of a tile state ('W', 'B', 'R', 'L', 'UW', ...)."""
3  qc = QuantumCircuit(1)
4  base = name[-1]
5  if base == "B":
6      qc.x(0)
7  elif base == "R":
8      qc.h(0)
9  elif base == "L":
10     qc.x(0)
11     qc.h(0)
12     if len(name) == 2:                                # the pink square, outermost
13         qc.ry(-2 * np.pi / 3, 0)
14     return qc

```

The pink square is $R_y(-2\pi/3)$ — the same matrix as equation (6), pinned entrywise in the notebook before use. All notebook experiments are seeded and reproducible; expected values are stated with full-precision literals and tolerances.

Important

Qiskit bit-ordering note. Qiskit orders multi-qubit bitstrings little-endian (rightmost bit = qubit q_0), while this lab series writes states big-endian (qubit 0 leftmost). Every entangled pair in QTris is a two-qubit register with qubit 0 = the lower-numbered cell, so *every* displayed pair outcome must be converted to the leftmost-first ordering of this guide. The notebook shows exactly how — a string reversal — and applies it to every histogram and state vector.

9.7 Activity 7: Quick Check

Quick Check

Question 1: A right-oriented bianconera is on cell 5. Which single card makes the upcoming colour measurement *certain*, and with which outcome?

- (a) X — certain black (b) H — certain white (c) Z — certain white

Question 2: A Ψ^+ pair (no markers) is measured. What is the probability that the two cells come up the *same* colour?

- (a) $1/2$ (b) 1 (c) 0

Question 3: $P(W)$ for a pink-decorated right-oriented bianconera is $\cos^2 15^\circ$. Which printed band encodes it, and how far is the print from the physics?

- (a) 1–75; off by 0.005 (b) 1–93; off by 0.003 (c) 1–93; exact

Question 4: Why can *no* choice of gate for the pink square make the printed double- U table right?

- (a) Because the printed Φ^+ and Φ^- rows together would force decorated colours to stay definite, contradicting the 25/75 row.
(b) Because two rotations can never cancel.

10 Resources

- Hamma, A.; De Simone, I.; Nazzaro, M.; Viscardi, M., *QTris e Meccanica Quantistica*, AH Quantum Lab / Kommunikart, online edition for the NQSTI site, 2024 (Italian). — The original game: rulebook and printable components; the normative source for every rule and printed table in this guide.
- NQSTI — National Quantum Science and Technology Institute, <https://www.nqsti.it>. — Publisher of the online edition and the ecosystem this adaptation belongs to.
- Rudolph, T., *Q is for Quantum*, 2017 (Italian translation *Quanti*, Adelphi, 2020 — the edition the rulebook recommends). — The black-and-white-ball pedagogy the game’s tiles descend from.
- Nielsen, M. A. & Chuang, I. L., *Quantum Computation and Quantum Information*, Cambridge University Press, 2010. — Pauli and Clifford algebra, Bell states, CNOT conventions.
- Seskir, Z. C. et al., “Quantum games and interactive tools for quantum technologies outreach and education,” *Optical Engineering* **61**(8), 081809 (2022). — Where games like QTris sit in quantum education.
- Lab 5: *Quantum Measurements*, Lab 6: *Quantum Key Distribution (BB84)* and Lab 7: *Quantum Optics* (this series). — Born rule, Procedures 1 and 2, mutual unbiasedness, and the pair-rotation invariance reused in the erratum section.
- Companion Jupyter notebook: `quantum-labs-2026/lab8/lab8_qtris.ipynb` (Colab link in Part 6).